

## Lab 4: The Formation of the Moon

The Moon was formed more than four billion years ago. For a long time, the exact mechanism by which we captured our moon wasn't exactly clear. The moon is odd – much bigger than most other moons when compared to the planet they are orbiting. In fact, out of the hundreds of moons orbiting planets in our solar system, only Pluto has managed to capture a relatively larger moon than the Earth has. The first goal of this lab is for your group to explore, and reason about which of four hypotheses about the moon is likely to be the correct interpretation. The possibilities are: (1) The Moon is a Sister Planet formed with the Earth, (2) the moon is a captured asteroid that wandered by (3) the moon is an Earth fragment that spun off (or "Fissioned") because of the incredibly fast spin rate of the Early earth, or (4) the Moon is the result of a large impact with another planet.

In the English language, there are many different styles of writing. Consider the difference in tone and substance of a Shakespearean Sonnet and an ad in a newspaper. For each kind of writing, there are certain features that are important. Scientific communication is no different. A second objective of this lab is to practice writing a formal lab report to communicate how your group thinks the moon formed.

### Learning Objectives

At the completion of this lab, you should be able to:

- Use the scientific method work out how the moon formed
- evaluate the strengths and weaknesses of four formation scenarios
- put your groups findings into a formal lab report
- hone your scientific writing skills

### The Structure of Lab Reports

Modern lab reports have evolved to have

The main goal of a lab report is to craft a narrative that: (1) motivates for the reader why the investigation was undertaken, (2) outlines the steps of the investigation in such a way that your experiment could be recreated, (3) presents the results of the investigation in an easy to understand way, and (4) grapples with the relevancy of the results relative to what the reader may expect.

- 1. The Introduction:** The goal of the introduction is to motivate the investigation for the reader. Basically – why should the reader care? As a general rule of thumb, the intro should be two paragraphs. Paragraph one frames the phenomenon being studied. Paragraph two explains the goals of the investigation.
- 2. The Procedure:** The goal of the procedure section is to outline the steps of your investigation so that it could be recreated should the reader choose to do so. Try to use an economy of words. What is the minimum number of steps and words that you could say to get the reader to accurately recreate your investigation? But be specific. The reader will need to know (1) what materials were used– even brand/make of scientific equipment can be important, (2) what the experimental setup was, and (3) the steps taken. Because we are using other people's data for procedure just say something like "**we are using data that has been obtained and validated by the scientific community over centuries.**" We'll practice writing good procedures later.

**3. The Results Section:** presents the results of your investigation. Here are some formatting ideas to help present your results in an accessible way:

- A. Plots and data tables should be presented in this section. Please include the minimum number of plots and figures necessary to fully document your findings.
- B. Plots need to be about 25% the width of the page, and should be appropriately scaled so that the data are as big as possible
- C. Think about stacking your data tables and plots to make your results easy to understand while using a minimum total amount of space. For example you could maybe include your data table and graph side-by-side to save space.
- D. Don't break data tables or plot series over multiple pages. Scale them down to fit snugly
- E. Label your axes and provide captions explaining what the data table or plot is. The reader needs to be able to infer all numerical (x and y values) of data.
- F. If you are fitting to the data please include all the parameters of the fit.
- G. You need to report some measure of statistical goodness of fit in order to frame how successful your results were.

**4. The Conclusion:** The purpose of the conclusions section is to try and assess how successful your results were in comparison to the goals of your investigation. In your case, the goal is to use evidence to support one of four possible formation scenarios of the moon.

If you fall short – maybe you can't decide why or how the moon formed based on the evidence provided, try to pinpoint why and think about what additional data would be needed to help you settle the debate.

The goal of this lab is for you to write a formal lab report, in a group, that has an introduction, procedure, results section, and conclusion. For this lab, it's likely that the intro and procedure will be quite short, while the results and conclusion sections will be much longer.

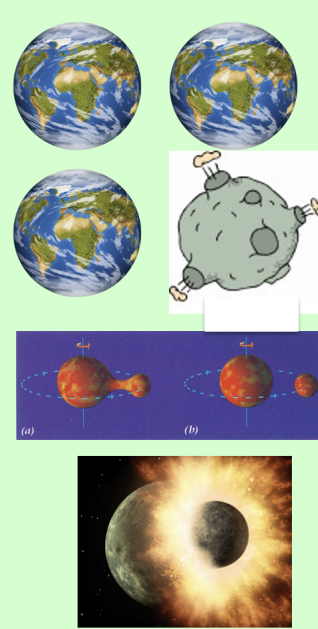
The lab has to be written out, and you will need to capture any images and embed them in your lab report to discuss them. Some recommendations for you.

- 1. Do not copy the text in the summaries of the formation hypotheses. Instead summarize them in your own words.
- 2. It may be fruitful to really think through what evidence would support each formation scenario, and then look for that evidence in the data given. It's especially helpful when one kind of evidence can differentiate between two scenarios.
- 3. Reformat any data tables you are inputting into your lab reports
- 4. You can put any actual plots or images into the lab report. Normally, you would obtain these data in a method outlined in your procedure. It's ok to use them for today, just cite them as (astroclass, 2024).

## Moon Formation Scenarios

There are four commonly discussed moon formation scenarios.

- 1. Sister Planet**  
An earth-like planet formed nearby
- 2. Captured Asteroid**  
Asteroid meandered by and was captured
- 3. Earth Fragment ("Fission")**  
Chunk flew off spinning Earth
- 4. Giant Impact**  
Mars sized planet collided with Earth



### 1. Sister Planet

Often called the "double planet hypothesis", this proposal is that the Earth and Moon formed at about the same time in the same region of the solar system and were close enough together to form a bound system with each other. It certainly seems like a reasonable hypothesis.

The first challenge to this hypothesis is the great difference in composition of the Earth and Moon. If they formed at about the same time in the same region of space by the same mechanisms, then they would be expected to be very similar in structure and composition. But the Earth has an average density of  $5.5 \text{ gm/cm}^3$  and the Moon has a much lower overall density,  $3.34 \text{ gm/cm}^3$ , indicating that it lacks the iron core. The densities of Mercury, Venus, and Mars are  $5.4 \text{ gm/cm}^3$ ,  $5.24 \text{ gm/cm}^3$ , and  $3.94 \text{ gm/cm}^3$  respectively. Even the majority of [meteorites](#) have densities greater than the Moon and many are mostly iron, suggesting that a large iron content is part of the expected composition of objects formed in the Earth's part of the solar system.

Arguments in favor of this hypothesis include the fact that the Moon's composition is very similar to material in the Earth's [mantle](#). Furthermore, the oxygen isotopic content of the Moon rocks is identical to that in Earth rocks and the [differentiated meteorites](#), suggesting that it formed in the same part of the solar system.

Attempts to resolve the problem of the composition differences between Earth and Moon include the proposal that the Moon accreted from a ring of planetesimals that formed around the Earth soon after its formation. A variation on this theme has the Earth capturing a small moon which then grew by accretion. But the Earth would presumably have grown by the same accretion process, and there does not seem to be a clear path to explaining the great difference in composition under the constraints of the double planet hypothesis.

## 2. Captured Asteroid

While advocates of this model differ in their proposed origin of the interloper, the fact that the [age of the Moon](#) is similar to the solar system makes it likely that the source of the material is from within the solar system. Dalrymple points out that the oxygen isotope concentration is identical in the Earth and Moon and the undifferentiated meteorites, but different from that in more primitive meteorites. This would seem to restrict the source of the Moon to the same general neighborhood of the solar system as the Earth, but the bulk composition of the Earth with its large iron core is greatly different from that of the Moon.

Can the Earth just capture a neighbor that wanders by and make a moon of it? Not without losing a lot of energy! A smaller body that encounters the gravity of a larger body will experience the [slingshot effect](#), more formally known as a gravity assist maneuver, and will escape the encounter traveling in another direction.

## 3. Fission Fragment

This scenario envisions the early Earth in its molten state spinning rapidly and forming a tidal bulge at the equator which led to the separation of the mass of the Moon and its coalescence as a separate body. Sometimes called the "fission" model, this hypothesis was put forward by George Darwin in 1879. Part of the model was the proposal that a resonance with the Sun and solar tides which gave the necessary extra boost to separate off a large molten mass which then solidified to form the Moon. In support of this hypothesis, Osmond Fischer proposed that the Pacific Ocean represents the unhealed scar where the Moon used to be. Further support came from the fact that the Moon's density is very nearly equal to the density of the Earth's [mantle](#).

The arguments against this hypothesis involve a number of interesting studies.

1. The Pacific Ocean basin is much too young to have been the Moon's exit point. [Radiometric dating](#) of the sea floor gives an age of about 200 million years compared to the projected [age of the Earth](#) at around  $4.5 \times 10^9$  years. Dating of the oldest Moon rocks gives an [age of the Moon](#) of about  $4.4 \times 10^9$  years.
2. "The [angular momentum](#) in the Earth-Moon system is insufficient by a factor of three for the proposed primordial Earth to have formed the tidal bulge necessary for fission." (Dalrymple) No reasonable mechanism has been found for the loss of enough angular momentum after separation to account for the deficit.
3. [Tidal friction](#) in the Earth would dampen a tidal bulge before it could become large enough to throw off part of its mass.
4. The "mass of the daughter planet produced by fission should have been about one fifth the mass of the Earth." (Dalrymple) No plausible mechanism can bridge the gap between one fifth of the Earth's mass and the observed mass of  $M_{\text{Earth}}/81.3$  for the Moon.
5. The newly formed Moon would be inside the [Roche limit](#) relative to the Earth and would be destroyed by tidal forces.
6. No plausible scenario would give the fission fragment (Moon) the [orbital velocity](#) necessary to remain in orbit about the Earth. Some matter might escape entirely, but most would fall back to Earth.
7. If the Moon were the result from the fission of an equatorial tidal bulge, then its plane of orbit should be that of the equator. But the Moon's orbit varies from  $18.5^\circ$  to  $28.5^\circ$ .

## 4. Giant Impact

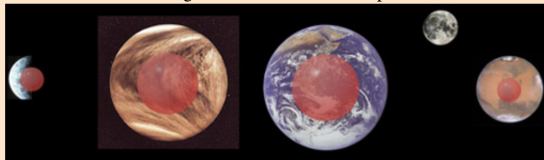
The compositions of the four inner "rocky" planets are similar with Mercury at density  $5.4 \text{ g/cm}^3$ , Venus at  $5.2$ , [Earth](#) at  $5.5$  and Mars at  $3.9 \text{ g/cm}^3$ . This is interpreted as evidence of a core of iron and nickel, with [Mars having a smaller core](#). Even the largest [meteorites](#) often show [iron/nickel cores](#), so that the iron core seems like a signature for the formation of massive objects in the inner part of the solar system.



But Earth's [Moon](#) with a density of  $3.3 \text{ g/cm}^3$  is very different from the Earth and from the other bodies in the inner solar system. This was one of the first clues that the origin of the Moon was not just accretion from the same material from which the Earth was formed.

Using data on the [structure of the Earth](#), a simplified perspective can be constructed of the inner planets and Earth's Moon. The size of Earth's core is shown, and the planet densities above are used to construct similar images of the other inner planets. All except the Moon have sizable iron cores. The [moons of Mars](#) are too small to show in this perspective.

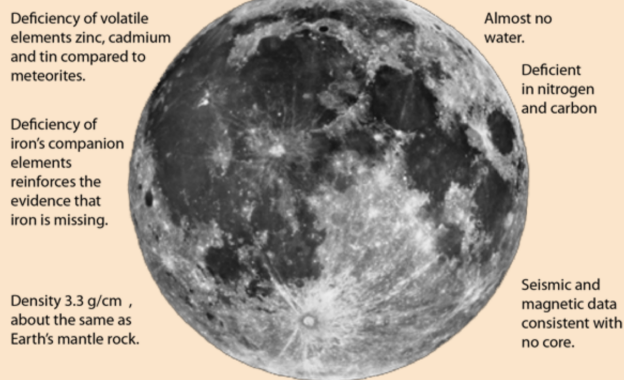
Modeling the iron cores of the inner planets.



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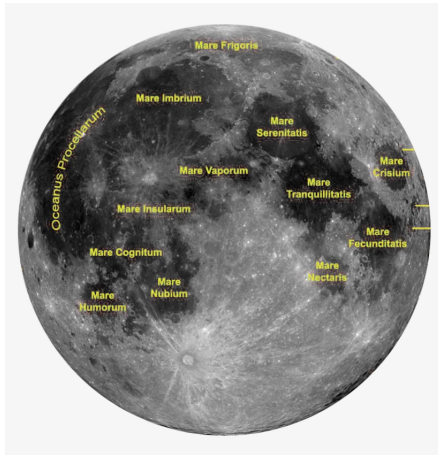
### Evidence for a hot, violent past for the Moon



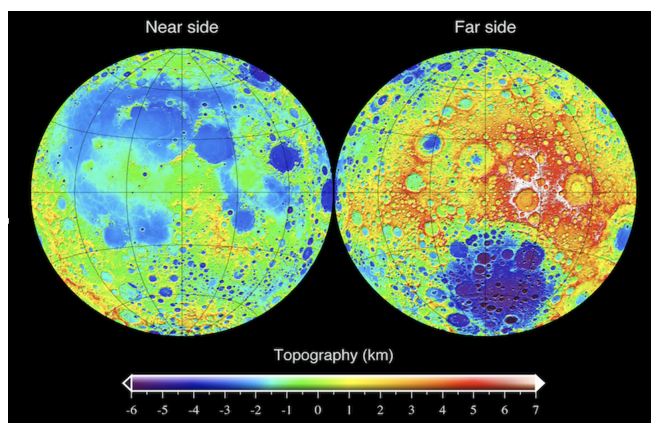
According to Ward and Brownlee, there is remarkable similarity between the trace element content of the Moon rocks and rocks from Earth's mantle. Also the isotopic content of the Earth and Moon samples was essentially identical.

## Data

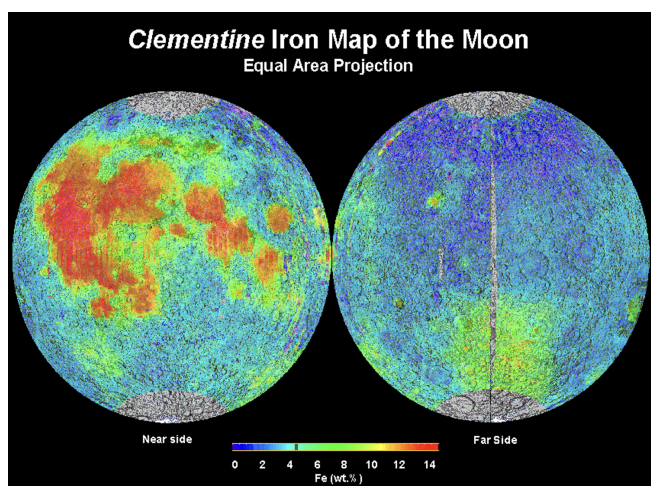
1. The moon has lowlands (**Mare** or Maria (pl.)) and highlands. The lowlands are made from lavalike (basaltic) material.



2. Topographical Asymmetry on the Moon



3. Unequal Iron Distribution



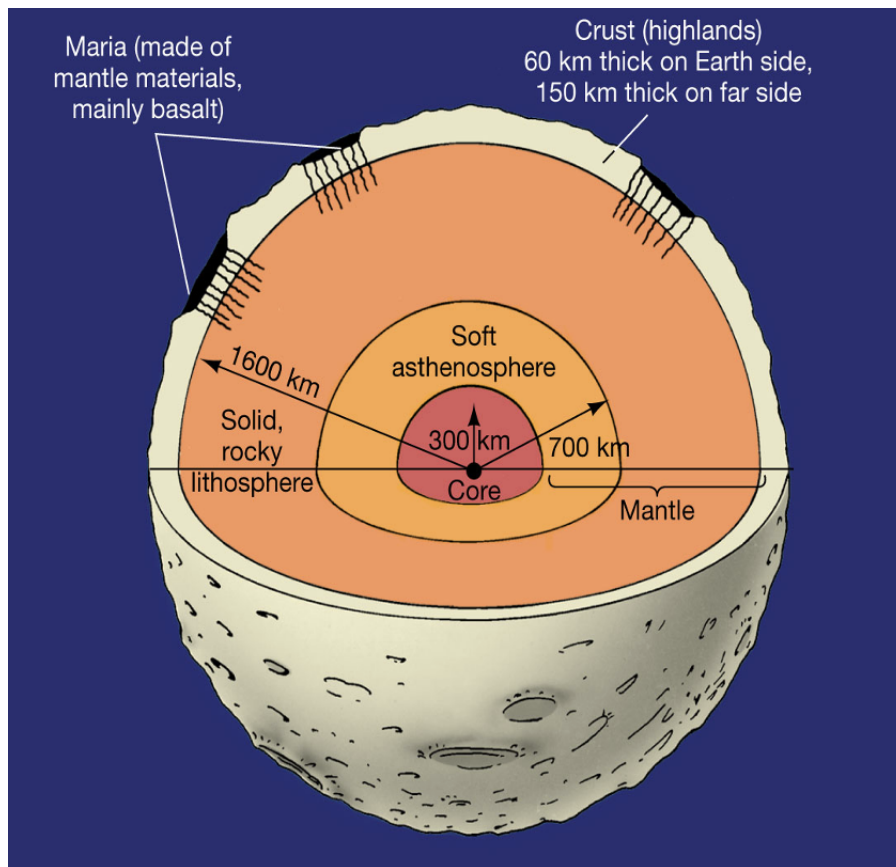
## 4. Elemental Abundances in Apollo Rock Samples

| Element | Earth Crust | Moon |
|---------|-------------|------|
| Fe      | 34.6        | 3.5  |
| O       | 29.5        | 60   |
| Mg      | 12.7        | 3.5  |
| Ti      | 0.05        | 1.0  |

Red = Siderophiles

Green=Lithophiles (from Lunar Highlands)

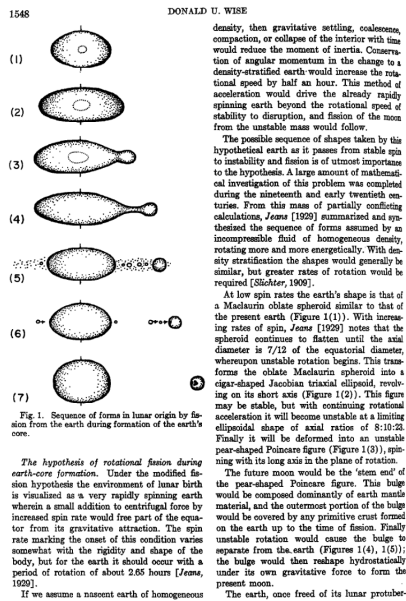
## 5. Model of Moon Interior



## 6. Terrestrial Planet Densities

| Object        | Density (g/cm <sup>2</sup> ) |
|---------------|------------------------------|
| Mercury       | 5.4                          |
| Venus         | 5.2                          |
| Earth         | 5.5                          |
| Earth's Crust | 3.3                          |
| Moon          | 3.3                          |

7. To fission off the Moon, the day would have to have been only 2.65 hrs. But we know it was longer than that.



## 8. Oxygen Isotopes from Apollo Samples

